

Talking-Head Generation in Practice

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Abstract

Talking-head generation has progressed rapidly in recent years, driven by advances in vision, speech, and generative modeling. Yet despite this momentum, the field lacks a clear, consolidated understanding of how its research themes, datasets, and evaluation practices have evolved. To address this gap, we curate a comprehensive corpus of over 100 influential works and derive a coherent semantic taxonomy that reveals the main directions shaping the area, including speech-to-motion representation, style- and emotion-aware animation, and high-fidelity diffusion models. Building on this taxonomy, we present the first longitudinal analysis of datasets and metrics used in talking-head generation from 2021 to 2025. Our findings uncover distinct trends: increasing dependence on audio-visual and emotion-rich datasets, and a rapid rise in newly proposed evaluation metrics, especially those targeting expression naturalness, audio-visual synchronization, and landmark accuracy or driving-signal alignment. These patterns indicate a shift in the community's priorities from frame-level realism toward semantic alignment, temporal coherence, and perceptual quality. By unifying methodological structure with quantitative historical insights, this survey offers concrete guidance for developing future talking-head systems, choosing appropriate benchmarks, and designing meaningful evaluation protocols. We expect the work to serve as a central reference for advancing expressive, controllable, and perceptually aligned talking-head generation. Our appendix is available [here](#).

CCS Concepts

• **General and reference** → **Surveys and overviews**; General literature; • **Computing methodologies** → *Machine learning*; *Computer vision tasks*.

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Keywords

Talking-head generation; Facial animation; Audio-visual modeling; Diffusion-based generation; Dataset analysis; Evaluation protocols; Semantic taxonomy

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1 Introduction

Talking-head generation [6, 18, 21, 23, 33–35, 45, 47, 66], synthesizing realistic and expressive human head motions from audio, text, or driving signals, has rapidly transformed from a niche problem into a central capability for digital humans, virtual communication, education, entertainment, and accessibility technologies. Fueled by advances in deep generative modeling, modern systems can now produce high-fidelity facial motion [59, 63], nuanced emotional expression [20, 40], and identity-preserving appearance from minimal inputs [31, 58, 62]. The last five years, in particular, have seen an explosion of new approaches spanning motion-representation learning [12, 53], neural radiance fields (NeRFs) [14, 59], diffusion-based synthesis [11, 40], and multimodal control [56]. This rapid progress has pushed the field forward at an unprecedented pace, but it has also made it increasingly challenging to understand the broader landscape.

Despite this momentum, several fundamental questions remain unanswered. *What core problem formulations are driving current research? How are the latest models different in terms of representation, controllability, and expressiveness? Which datasets and evaluation protocols are shaping the community's standards, and how have they changed over time? As new models emphasize emotion, style, or semantic alignment, are existing metrics still adequate? And most importantly, what emerging trends reveal where the field is heading next?* The literature has grown too quickly and too heterogeneously for these questions to be answered through anecdotal understanding alone [32, 46, 60].

Existing surveys and tutorials [10, 19, 38, 41, 65] cover only narrow subsets of talking-head generation, e.g., lip synchronization, animation frameworks, or specific model families, and thus do not

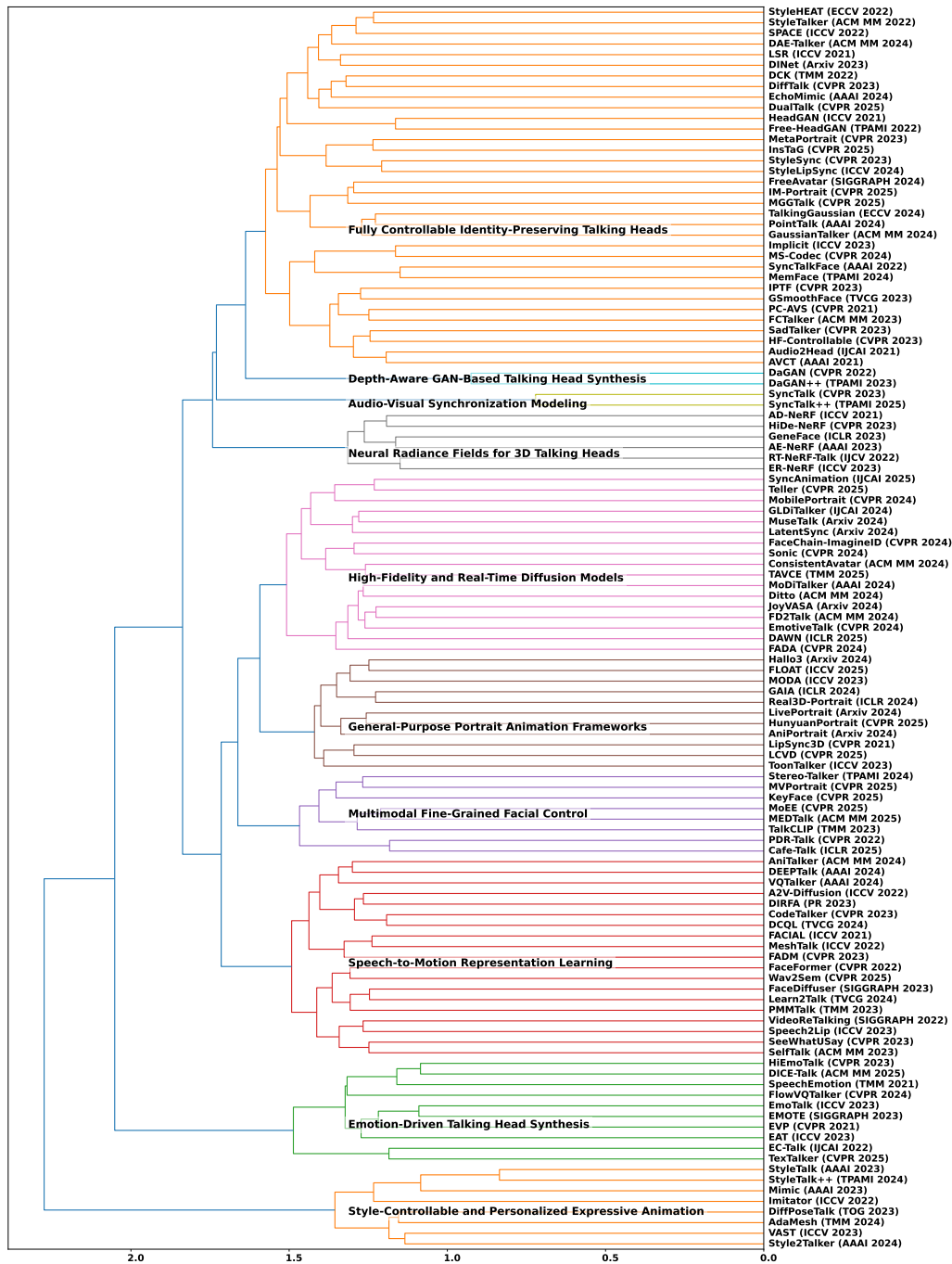


Figure 1: Hierarchical semantic taxonomy of 117 representative talking-head generation methods from 2021-2025. The taxonomy shows ten major research directions that have naturally emerged from data-driven clustering of paper abstracts and titles, spanning foundational advances, such as speech-to-motion representation learning, NeRF-based 3D heads, and high-fidelity diffusion models, to higher-level themes in style control, emotion modeling, and multimodal fine-grained facial control. Adjacent clusters highlight areas of conceptual convergence, including the rise of expressive, personalized, and identity-preserving generation. The horizontal axis represents the linkage distance in hierarchical clustering, indicating the degree of semantic dissimilarity between groups of methods. Collectively, the taxonomy offers a structured map of the field's evolution, illustrating how modern talking-head research is diversifying from low-level realism toward semantic alignment, controllability, and fully generalizable portrait animation frameworks.

capture the breadth, diversity, and evolution of the field. Meanwhile, experimental practices have shifted dramatically: datasets have expanded from celebrity video corpora [7, 30] to emotion-rich [25], high-resolution [64], and multi-view collections [50]; metrics have diversified from simple pixel-space measures [16, 51] to those evaluating semantic correctness [5, 13], audio-visual coherence [33, 57], and driving-signal alignment [53]. Yet there is currently no systematic, quantitative analysis showing how these choices have evolved and what they imply for future research.

To address these gaps, we curate a comprehensive corpus of more than 100 influential papers from 2021–2025 and analyze them through both semantic structure and experimental trends. Our goal is not only to map the methodological landscape but also to provide practical guidance for designing future systems, selecting appropriate benchmarks, and developing meaningful evaluation standards. The resulting survey aims to offer a clear, consolidated view of a rapidly evolving research area, and to serve as a foundation for the next generation of expressive, controllable, and perceptually aligned talking-head models. Our **main contributions** are threefold:

- i. A unified taxonomy of talking-head generation, derived from a curated corpus of over 100 influential papers and revealing the major research directions that define the current landscape.
- ii. The first longitudinal analysis of datasets and evaluation metrics (2021–2025), uncovering trends in dataset selection, benchmark evolution, and the community’s shifting focus toward semantic, temporal, and perceptual evaluation.
- iii. Actionable guidance for future research, highlighting experimental best practices, underexplored problem settings, and emerging opportunities for building more expressive, controllable, and reliable talking-head generation systems.

We review existing surveys and studies in Appendix; below, we present a unified perspective on this field from several aspects.

2 Framework for Unified Analysis

Appendix provides full details of the corpus curation process, semantic taxonomy methodology, and experiment extraction pipeline. Below we present our analytical framework and detailed analysis.

2.1 Analytical Framework

Our analytical framework is built on a curated corpus of 117 representative talking-head generation papers, selected through a multi-stage process designed to balance completeness and quality.

We begin with broad keyword-based retrieval from arXiv, then enrich each entry with metadata and code links. Peer-reviewed publications are automatically retained, while arXiv-only preprints are filtered using community impact signals, ensuring inclusion only when the work has demonstrable influence (e.g., implementation popularity). A final semantic screening removes papers that fall outside the scope of explicit talking-head generation. This yields a corpus that captures the major model families (e.g., GANs, NeRFs, diffusion models, transformers, *etc.*), the dominant control modalities (e.g., audio-driven, text-driven, motion-driven, *etc.*), and the community’s most visible and methodologically mature contributions. From this corpus, we extract all dataset and evaluation-metric information by parsing each paper’s experiment section and processing it with a structured LLM-based extraction prompt. Dataset

names, metric names, and training/testing splits are normalized to unified identifiers, enabling consistent cross-paper comparison. These standardized elements form the basis for our longitudinal analyses of dataset usage and metric evolution.

To reveal the thematic structure of the field, we construct a hierarchical semantic taxonomy using TF-IDF [39] representations of titles and abstracts, followed by hierarchical agglomerative clustering with Ward’s linkage. This method is chosen because it minimizes within-cluster variance and naturally exposes multi-level topic organization. The resulting dendrogram provides a visual and data-driven map of research themes, while cluster labels are refined using a large language model to ensure interpretability.

This pipeline (e.g., representative corpus construction, structured experiment extraction, and hierarchical semantic mapping) forms a coherent framework that supports all subsequent analyses of trends, practices, and emerging directions in talking-head generation.

2.2 Semantic Taxonomy of Research Directions

Figure 1 presents the hierarchical semantic taxonomy derived from our corpus of 117 representative works, offering a view of how the talking-head generation community has organically structured itself over the past five years. It provides not just a static categorization, but a conceptual map that shows how the field has evolved, where current momentum is concentrated, and which thematic connections may shape the next generation of talking-head systems. Rather than imposing predefined categories, the taxonomy emerges directly from data-driven clustering, showing ten coherent research directions that together map the conceptual terrain of the field.

At the top of the taxonomy lie methodological backbones, such as speech-to-motion representation learning, NeRFs for 3D talking heads, and high-fidelity diffusion models. These clusters represent the generative foundations underpinning most recent advances. Their clear separation highlights how the community has diversified beyond traditional encoder-decoder pipelines: NeRF-based 3D heads emphasize spatial consistency and view-dependence; diffusion models drive improvements in realism and temporal stability; and representation-learning methods focus on robust, speaker-independent motion modeling. The size and density of these clusters indicate that they collectively power much of the field’s technical momentum. A second set of clusters reflects emerging forms of control and expressiveness, including style-controllable and personalized animation, emotion-driven synthesis, multimodal fine-grained facial control, and audio-visual synchronization modeling. These areas illustrate a shift from lip-sync correctness toward richer communicative behaviors, emotional nuance, stylistic identity, controllable expressivity, and fine-grained semantic responsiveness. Their close spatial proximity in the taxonomy shows that many recent works operate at the intersection of these goals, blending speech cues, emotional priors, and multimodal signals to achieve more lifelike head motion.

Finally, two clusters, general-purpose portrait animation frameworks and fully controllable identity-preserving talking heads, capture systems that integrate multiple capabilities into unified frameworks. These works often pull techniques from several upstream clusters, reflecting a trend toward versatile pipelines that

support reenactment, instruction-based generation, long-term identity preservation, and user-controllable behavior. The prominent position of these clusters suggests that the community increasingly values practical, deployable solutions that can generalize across identities, styles, and input modalities.

The taxonomy illustrates three key insights. First, the field has become structurally multidimensional: progress is no longer driven by a single innovation path, but by parallel advances across representation, controllability, realism, and user alignment. Second, there is a convergence toward models that reconcile expressiveness with reliability, seen in the adjacency of emotional, stylistic, and identity-preserving clusters. Third, the boundaries between categories are porous: many recent systems sit at the intersections, indicating a maturation of the field toward holistic, multimodal, and semantically grounded talking-head generation.

2.3 Longitudinal Trends in Dataset Usage

Figure 2 summarizes the longitudinal trends in dataset selection, grouped into four major categories, face image datasets, audio datasets, audio-visual datasets, and emotion-rich datasets, while distinguishing training and testing usage. The figure exposes a clear transition in the data foundations of talking-head research, reflecting the community's expanding requirements for expressiveness, multimodal alignment, and performance generalization.

In the early years, the landscape is dominated by audio-visual video datasets, particularly VoxCeleb1/2 [7, 30], LRS2/3 [1, 2], and GRID [9]. These datasets provided large-scale, speaker-diverse audio-visual pairs, making them ideal for traditional lip-sync and audio-driven head-motion models. Their consistent presence across both training and testing indicates their role as de facto benchmarks during this period. However, their relative dominance gradually declines as more specialized datasets begin to enter the ecosystem. From 2023 onward, we observe a rising dependence on high-resolution and multi-view portrait datasets, as seen in the growing use of CelebV-HQ [67], HDTF [64], and other sources. This trend mirrors the shift toward 3D-aware and diffusion-based models (see Figure 1), which require richer spatial variation, higher fidelity appearances, and more consistent lighting conditions to train view-consistent head renderers and NeRF-style representations.

A particularly notable trend is the steady increase in emotion datasets used for both training and evaluation. Starting from almost negligible usage in 2021, emotion datasets, such as CREMA-D [4] and RAVDESS [25], become substantially more common in later years, especially by 2024-2025. This growth directly reflects the community's expanding interest in expressive and emotionally aligned talking-head generation, where models should move beyond neutral lip movement to capture affective nuance, speaking style, and conversational authenticity. Meanwhile, audio-only datasets maintain consistent but modest usage, primarily serving as supplementary resources when large-scale or domain-matched speech is required. Their secondary role underscores that talking-head generation fundamentally relies on synchronized audio-visual data or emotion-annotated corpora, rather than speech alone.

Across all categories, the increasing diversity of training-testing splits highlights a broader methodological maturation: models are

tested on data distributions increasingly different from those they were trained on. This reflects growing emphasis on generalization, robustness, and evaluation fairness, central concerns as talking-head technologies move from controlled laboratory conditions to real-world applications.

This shows an evolution from early reliance on generic, large-scale audio-visual datasets toward high-resolution, emotion-rich, multi-view, and semantically annotated datasets. This shift demonstrates that the field is no longer satisfied with generating visually plausible motion, but is now targeting identity robustness, expressive fidelity, and semantically coherent behavior, all of which demand more diverse and specialized data foundations.

2.4 Evolution of Evaluation Metrics

Evaluation metric landscape. Figure 3 presents a longitudinal analysis of evaluation metrics used in talking-head generation, organized into five core metric families: visual quality, audio-visual synchronization, geometry and landmark accuracy, expression naturalness, and driving-signal alignment. The figure shows a notable shift in how the community measures progress, reflecting deeper changes in research priorities and modeling capabilities.

Early in the timeline, research is dominated by visual-quality metrics, e.g., SSIM [51], PSNR [16], LPIPS [43], and FID [15]. These metrics, inherited from general video synthesis, were sufficient when the primary challenge was achieving realistic textures, stable frames, and identity preservation. However, their relative decline after 2023 indicates a growing consensus that frame-level similarity is no longer the main bottleneck for state-of-the-art systems.

Starting in 2023, we observe a steady rise in audio-visual synchronization metrics (e.g., LSE-C/D [8], Sync-C/D [33]), reflecting renewed attention to temporal coherence and phoneme-viseme alignment as models achieve higher realism but still struggle with precise timing. This trend underscores the community's recognition that perceptual synchronization is essential for speech-driven animation and profoundly impacts user trust and perceived naturalness. Even more striking is the rapid growth of geometry and landmark accuracy metrics and expression naturalness metrics. Metrics such as LMD, F-LMD, emotion accuracy, and smoothness [24, 29, 33, 42, 54] become increasingly common after 2023, peaking around 2024-2025. This shift shows a deeper transition in the field: from surface-level realism to semantic correctness, evaluating whether the generated facial motion matches linguistic cues, emotional intent, and human expressive norms. These metrics reflect the rising influence of emotionally aware and style-controlled models, which require more nuanced evaluation beyond pixel fidelity.

Driving alignment metrics (e.g., APD, AKD, ARD) [18, 22, 61], see meaningful adoption closer to 2025. Their growth suggests an emerging focus on controllability, where the goal is not merely to produce realistic motion but to ensure that user-specified control signals, motion trajectories, landmarks, expression codes, are faithfully reproduced. These metrics are critical for next-generation systems emphasizing fine-grained, user-driven manipulation.

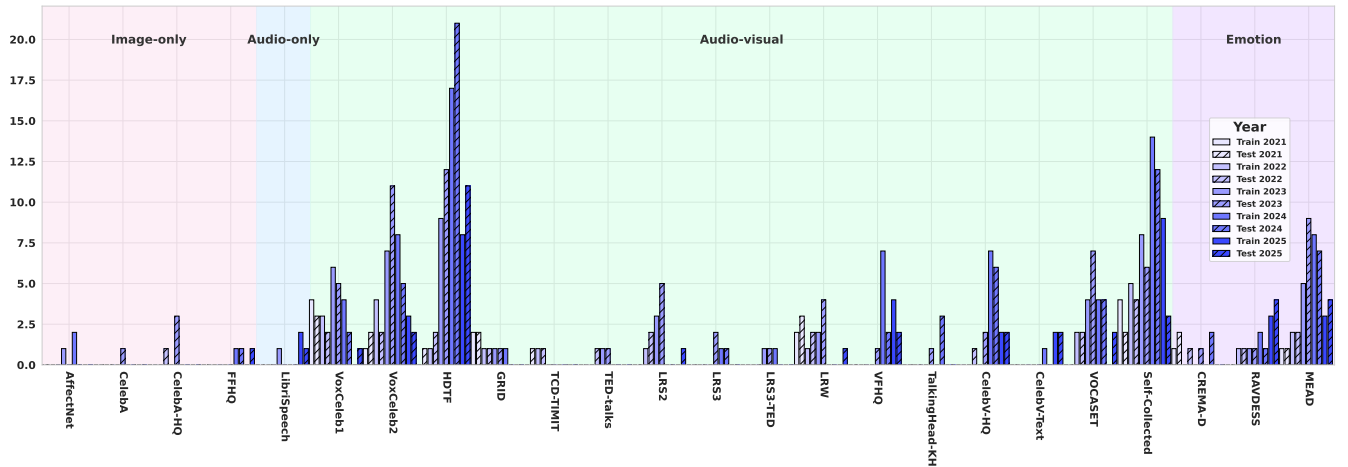


Figure 2: Longitudinal trends in dataset usage. Datasets are grouped into four categories, face images, audio-only speech datasets, audio-visual datasets, and emotion-rich datasets, separately tracked for training and testing. The vertical axis represents the number of papers that use the corresponding dataset in a given year. The figure highlights a shift from early reliance on large-scale audio-visual benchmarks (e.g., VoxCeleb, LRS, etc.) toward growing adoption of high-resolution portrait, multi-view, and emotion-annotated datasets (e.g., HDTF, MEAD, RAVDESS, etc.). This evolution reflects the field’s increasing emphasis on expressiveness, semantic alignment, and generalization beyond standard lip-sync benchmarks.

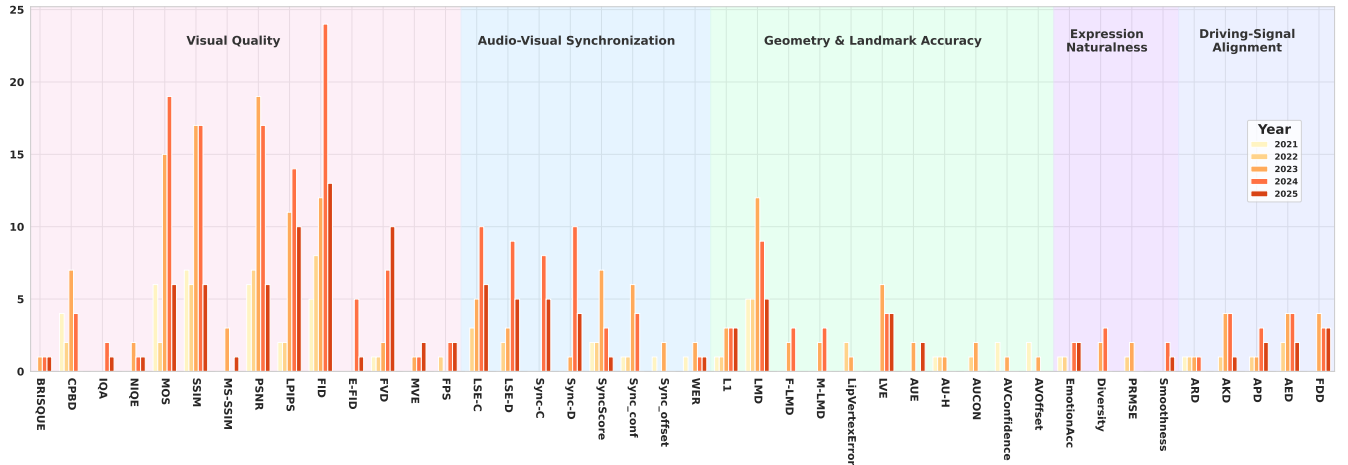


Figure 3: Longitudinal evolution of evaluation metrics used. Metrics are grouped into five functional categories: visual quality, audio-visual synchronization, geometry/landmark accuracy, expression naturalness, and driving-signal alignment. The vertical axis represents the number of papers that use the corresponding evaluation metric in a given year. The trends show a clear shift from early reliance on pixel-level fidelity measures toward richer, semantically oriented evaluation, including synchronization accuracy, emotional expressiveness, and controllability. This progression highlights the community’s move toward assessing temporal coherence, semantic alignment, and expressive behavior rather than surface-level realism alone.

Our analysis shows a clear trajectory: the field is maturing from evaluating how real the frames look to how semantically correct, expressive, and controllable the generated motion is. The increasing diversity of metrics also reflects an expanding understanding of talking-head generation as a multimodal communication task, not just a visual synthesis problem. This evolution highlights the need for more unified, perceptually grounded metrics and suggests that future progress will depend on evaluation protocols capable of capturing high-level behaviors, e.g., emotional resonance, conversational authenticity, and user-intent alignment.

Shifts in metric priorities over time. Figure 4 provides a two-part view of how evaluation practices in talking-head generation have evolved across 117 papers. Figure 4a visualizes the usage frequency of the five major metric families each year, while Figure 4b shows how newly introduced metrics are distributed across these same categories. They show not only what the community measures, but how its priorities and innovation efforts have shifted as models have progressed from basic lip-sync generation to expressive, controllable, and multimodal talking-head systems.

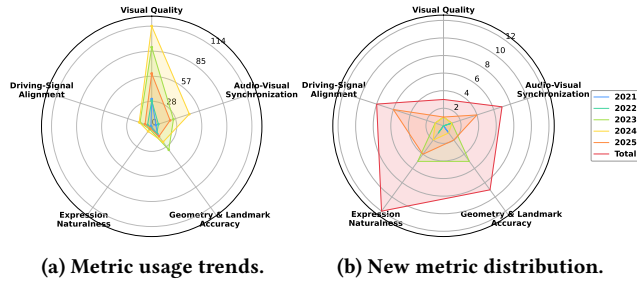


Figure 4: Evolution of evaluation metrics. Both subfigures share the same legend. (a) Radar visualization of metric usage across five major dimensions, showing the field’s early dependence on visual-quality measures and its gradual shift toward audio-visual synchronization and geometric accuracy as models pursue greater temporal coherence and semantic correctness. (b) Distribution of newly proposed evaluation metrics over time, highlighting a clear rise in measures targeting expression naturalness, driving-signal alignment, and fine-grained motion accuracy.

The left radar plot (Figure 4a) shows that visual-quality metrics (e.g., LPIPS, SSIM, PSNR) dominated early years, reflecting a stage where improving sharpness, identity fidelity, and artifact reduction was the foremost challenge. Their relative share decreases over time, not because they disappear, but because progress in generative backbones (GANs $\rightarrow$ NeRFs $\rightarrow$ diffusion models) has reduced the centrality of pixel-level fidelity as the primary bottleneck. Meanwhile, the remaining four dimensions: audio-visual synchronization, geometry and landmark accuracy, expression naturalness, and driving-signal alignment, all expand steadily, indicating a broadened understanding of what constitutes good talking-head generation. Synchronization metrics grow notably after 2023, aligned with audio-aware diffusion models and transformer-based speech, motion encoders that require precise temporal consistency. Geometry-related metrics increase in step with the rise of 3D-aware and NeRF-based heads, where spatial correctness and stable structure matter as much as appearance.

The right radar plot (Figure 4b) shows the direction of metric innovation. Newly proposed metrics concentrate not in visual quality, where mature measures already exist, but in expression naturalness, landmark/geometry accuracy, and driving-signal alignment. This indicates that researchers are actively developing tools to evaluate behavioral and semantic performance: emotional expressiveness, physically plausible motion, responsiveness to control signals, and fine-grained alignment with speech or driving inputs. Synchronization-related metrics also appear among new proposals, reflecting a renewed focus on perceptual audio-motion coherence.

The combined evidence from Figures 4a and 4b demonstrates a clear maturation of evaluation methodology. The field has moved beyond asking “Does the face look realistic?” toward “Does the face behave realistically, respond correctly, and communicate meaningfully?”, a shift that mirrors the trajectory of modern talking-head research toward expressiveness, controllability, multimodal grounding, and human-aligned communication.

3 Findings and Emerging Trends

The analyses presented above collectively show a field that is not only expanding in capability but also maturing in scope and ambition. In this section, we synthesize these findings across three dimensions, models, datasets, and metrics, to highlight emerging trends, persistent challenges, and future opportunities.

3.1 Evolution of Model Paradigms

The semantic taxonomy in Figure 1 illustrates how the field has diversified into multiple coherent yet interconnected research directions. Early audio-driven models, which focused primarily on phoneme-viseme alignment and frame-level realism, have now been complemented, and in some cases surpassed, by multimodal and highly expressive approaches.

Foundational modeling innovations. Three clusters, speech-to-motion representation learning, NeRF-based 3D heads, and high-fidelity diffusion models, represent the core methodological engines driving recent progress. Representation-learning methods emphasize generalizable motion priors, NeRF-based systems elevate spatial realism and view consistency, and diffusion models substantially boost visual fidelity and temporal stability.

Shift toward expressiveness and alignment. Clusters related to emotion-driven synthesis, style control, and fine-grained multimodal manipulation demonstrate a clear pivot from lip-sync correctness to richer communicative behavior. Models introduced from 2023 onward increasingly incorporate emotion cues, prosody, semantic context, or user instructions, resulting in head motions that exhibit personality, style, and narrative coherence.

Toward unified & controllable systems. The emergence of general-purpose animation frameworks and fully controllable identity-preserving systems suggests a trend toward integrated solutions capable of reenactment, audio-driven generation, style modulation, and semantic alignment within a single architecture. These systems reflect the field’s ambition to support real-world deployment, where controllability, rather than raw realism, is decisive.

These model trends show a clear trajectory: talking-head generation is evolving from narrowly defined, modality-specific synthesis tasks into a broader paradigm of expressive, controllable, and semantically grounded digital human communication.

3.2 Transformation of Dataset Foundations

Figure 2 shows a parallel evolution in dataset usage. Early models depended heavily on large-scale audio-visual datasets such as VoxCeleb. These datasets enabled progress in lip synchronization and basic head motion generation but provided limited coverage of emotion, style, or high-frequency appearance details.

Rise of high-fidelity and multi-view datasets. As NeRF-based and diffusion-based models gained popularity, the reliance on high-resolution and multi-view datasets increased. Resources such as CelebV-HQ, HDTF, and other high-quality portrait collections provide the spatial richness needed for geometric consistency and fine-grained appearance modeling.

Growing importance of emotion and expressive datasets. One of the strongest trends after 2023 is the increased use of emotion-rich and affect-annotated datasets (e.g., RAVDESS, MEAD,

and self-collected). These datasets support the modeling of expressive cues, an area that traditional lip-sync corpora cannot cover. Their rising adoption also reflects the community’s recognition that expressive realism matters as much as pixel-level realism.

Training-testing divergence as a mark of maturity. As the field matured, models increasingly trained on specialized datasets while testing on diverse or unseen distributions. This shift indicates a push toward robustness, generalization, and fair benchmarking, all of which are essential for real-world applications.

Dataset evolution mirrors model evolution: to generate expressive, controllable talking heads, the field is gradually adopting datasets that are richer in emotion, higher in resolution, and more diverse in viewing conditions.

3.3 Maturation of Evaluation Practices

Figures 2-4 collectively show a decisive evolution in how talking-head generation is evaluated. The shift from traditional visual metrics to semantically grounded measures aligns with the field’s conceptual transformation.

Pixel-level visual metrics. SSIM, PSNR, and FID dominated early evaluation but gradually lost prominence as models reached high visual fidelity. These metrics were never sufficient to capture temporal coherence, emotional intent, or semantic correctness, and their decline reflects increased awareness of these limitations.

Rise of audio-visual synchronization metrics. Metrics such as LSE-C/D and SyncScore surged beginning in 2023, corresponding with growing emphasis on natural timing, speech-driven dynamics, and prosody-aware motion. Synchronization is now seen as a key determinant of perceived naturalness and trustworthiness.

Geometry and landmark accuracy. With the proliferation of 3D-aware and deformation-based models, landmark- and geometry-based metrics (e.g., LMD, F-LMD) have become essential for evaluating structural correctness. Their adoption signals that models are now judged not only by appearance but by their ability to maintain identity, pose accuracy, and geometric consistency.

Emergence of expression and driving-alignment metrics. The steep rise in expression naturalness and driving-alignment metrics after 2023 is perhaps the most important trend. As models incorporate emotion, style, and multimodal control, evaluation should consider whether generated motion reflects intended affect, follows control signals, and behaves consistently over time.

This metric evolution reflects a deeper philosophical shift: talking-head generation is no longer evaluated solely as a graphics problem, but increasingly as a communication problem, where semantics, timing, and expressiveness matter as much as pixel quality.

3.4 Practical Guidelines

Below we provide some practical guidelines.

We recommend adopting a compact, standardized metric suite that balances perceptual, temporal, geometric, and controllability signals. For perceptual quality report LPIPS and FID (or E-FID for expressive/affective domains) together with a frame-level visual fidelity measure (e.g., PSNR). For synchronization report an audio-visual sync metric such as LSE-C/LSE-D or SyncScore and include

Sync-offset statistics. For geometry and identity preservation report landmark/vertex errors (LMD or LipVertexError) and an identity similarity score (e.g., ArcFace). For expression and affect fidelity report AU consistency or emotion-classification accuracy on emotion-annotated test sets. For controllability report driver-alignment measures (AKD/APD/ARD) and a smoothness/stability metric (e.g., velocity/acceleration MSE). Automatic evaluation should be complemented by human perceptual judgments, e.g., run a small MOS study with at least 30 independent raters per method and report mean MOS with 95% confidence intervals. Compute metrics per-video and aggregate results using medians and interquartile ranges (and means where appropriate), and release evaluation scripts and exact hyperparameters to ensure reproducibility.

For test splits we suggest a multi-faceted protocol to stress generalization, expressiveness, and controllability. Use (i) an IID holdout by withholding 20% of speakers from the primary training corpus to measure in-distribution performance; (ii) a cross-dataset split that evaluates models on a high-fidelity portrait dataset (e.g., CelebV-HQ or HDTF) unseen during training; (iii) an emotion/expressiveness split using emotion-annotated corpora (e.g., CREMA-D, RAVDESS) to measure affective fidelity; and (iv) a low-shot personalization and novel-view split (1 - 5 frames for personalization and a multi-view/novel-view dataset such as HDTF) to evaluate identity stability and view consistency. For each split report per-condition statistics (gender, age, lighting) where available and pair automatic metrics with the MOS study described above. Finally, publish the exact split definitions, sampling seeds, and all evaluation code to enable fair, reproducible comparison.

3.5 Future Outlook

The evolution of talking-head generation from 2021 to 2025 reveals a field transitioning from narrow, task-specific pipelines to a broader paradigm of expressive, controllable, and semantically aligned digital human communication. By integrating insights from our taxonomy, dataset analysis, and evaluation trends, a coherent picture emerges of how the community has expanded its technical foundations and redefined its ambitions (see also Appendix).

From a modeling perspective, the field has moved decisively beyond traditional 2D appearance-based synthesis. Recent systems increasingly incorporate 3D-aware representations, multimodal encoders, and diffusion-based refinement modules that jointly enhance geometric consistency, expressive motion, and temporal stability. This progression reflects a shift from focusing on lip-sync realism alone to modeling communicative behavior more holistically, including emotion, prosody, speaking style, and semantic cues. A growing emphasis on controllability further underscores this shift. Modern architectures aim not only to generate photorealistic faces but also to expose interpretable control interfaces through landmarks, expression codes, style tokens, or textual instructions, enabling fine-grained and user-driven animation.

These changes in modeling practice have been accompanied by equally notable shifts in dataset usage. Early work relied heavily on large audio-visual corpora such as VoxCeleb, which remain valuable but are insufficient for capturing emotional nuance or high-quality geometric detail. The field now increasingly depends on high-resolution, multi-view, and emotion-rich datasets, which

offer the diversity and expressive coverage needed for training models that mimic natural conversational behavior. The trend toward training on specialized datasets while testing on diverse or unseen sets also signals a maturing understanding of generalization and robustness, which are essential for real-world deployment.

Evaluation practices have matured in parallel. Traditional visual-fidelity metrics, *e.g.*, SSIM, PSNR, or FID, have gradually lost dominance as researchers recognize their limitations in capturing temporal coherence or expressive realism. Synchronization measures, geometric and landmark accuracy metrics, and expression-oriented criteria have become central to evaluation, reflecting an understanding that speaking heads must be assessed not solely by their appearance but by the behavioral correctness and perceptual plausibility of their motion. This shift from pixel-level to semantic and perceptual evaluation highlights a broader reorientation of the field toward modeling communication rather than visual frames alone.

These developments point toward several promising opportunities. Semantic grounding and conversational consistency remain underexplored, and deeper integration of linguistic context could yield more coherent and contextually appropriate motion. Personalization at extremely low data cost is another challenge, as current approaches still require substantial footage to capture identity and expressive priors. Ethical considerations, including misuse prevention, watermarking, consent mechanisms, and bias mitigation, will become increasingly important as realism continues to improve. Finally, a longer-term frontier lies in unified generative humans that integrate head motion, gaze behavior, emotional state, and upper-body gesturing into a single cohesive model.

Talking-head generation is entering a new phase where semantic correctness, expressive fidelity, controllability, and reliability are becoming as important as visual realism. The field's trajectory suggests that future progress will depend on jointly advancing model architectures, dataset design, evaluation protocols, and ethical frameworks. These directions offer practical guidance for researchers and practitioners seeking to build the next generation of expressive, trustworthy, and context-aware digital humans.

4 Ethics, Limitations, and Conclusion

Ethical considerations. Talking-head generation technologies carry significant ethical implications due to their potential for both beneficial and harmful use. While recent advances enable more natural, expressive, and controllable digital humans, the same capabilities can be misappropriated for impersonation, misinformation, harassment, and other malicious applications [3, 26, 27, 36, 49, 52, 68].

Many datasets used in this field contain real individuals whose images and voices may not have been collected with explicit consent for synthetic content generation, raising concerns about privacy, consent, and downstream data reuse.

In addition, demographic imbalances in commonly used training corpora may amplify biases in identity preservation, expressive fidelity, and emotional rendering, leading to unequal performance across gender, age, or ethnic groups [28, 48, 55]. To mitigate such risks, researchers should adopt responsible data handling practices, use consent-aware or licensed datasets whenever possible, report

demographic composition transparently, and consider integrating watermarking or provenance signals into generative pipelines.

We recommend that future work include explicit discussions of intended use, potential harms, and safety constraints, and that evaluations incorporate robustness and misuse-resilience criteria. Addressing these ethical and societal issues is essential for ensuring that advances in talking-head generation contribute positively to human communication rather than eroding trust or enabling deceptive media [17, 37, 44].

Limitations. Although the analysis in this survey draws on a large and diverse corpus of 117 papers, several limitations may influence the completeness and generality of our findings. First, the paper selection is based on keyword-driven searches and public availability, which may bias the corpus toward popular research directions. Second, while our taxonomy construction combines TF-IDF clustering with LLM-assisted labeling, both steps introduce methodological sensitivity: clustering outcomes depend on vocabulary choices and similarity thresholds, and LLM summaries may reflect prompt phrasing or model priors. Third, the extraction of dataset and metric usage is subject to inconsistencies in how papers report their experimental setups, and despite manual verification, occasional omissions or misclassifications may remain. Fourth, our longitudinal trends reflect the publication distribution of 2021–2025 and should not be interpreted as causal evidence of methodological superiority. Finally, because our survey focuses on publicly documented methods, it cannot fully capture proprietary datasets, unpublished internal benchmarks, or emergent industrial practices. These limitations do not undermine the core insights but highlight the need for continued community curation, open artifacts, and standardized reporting protocols.

Conclusion. Our findings reveal a field that has rapidly progressed from frame-level lip-sync synthesis toward expressive, controllable, and semantically grounded head motion generation. Modern systems increasingly use 3D-aware representations, diffusion models, and multimodal control signals to achieve higher fidelity, stronger identity preservation, and more natural communicative behavior. In parallel, we observe a clear shift toward richer and more diverse datasets, including high-resolution, multi-view, and emotion-annotated corpora, and toward evaluation metrics that emphasize perceptual realism, temporal coherence, geometric correctness, and expressive fidelity. These developments reflect a growing recognition that talking-head generation should be assessed not only by visual similarity but also by its alignment with linguistic, emotional, and user-provided cues. The trajectory of the field points toward digital humans that are more trustworthy, expressive, and contextually aligned. We hope that the taxonomy, analyses, and insights presented here provide a useful foundation for future research, guiding the development of more controllable, perceptually aligned, and ethically responsible talking-head generation systems.

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